



University
of Windsor

Comp 3150 (60-315) Winter 2020

Final Examination (For 2026 Practice)

Examiner: Dr. C. I. Ezeife:

Given: Thurs., Apr. 9, 2020

Student Name: _____

Student Number: _____

INSTRUCTIONS (Please Read Carefully)

Examination Period is 2 hours 30 minutes

Answer all questions. Write your answers in the spaces provided in the question paper. This is closed book and closed notes test. **WRITE IN PEN or TYPE IN.**

Total Marks = 100. Total number of sections = 2

Please read questions carefully! Misinterpreting a question intentionally or unintentionally results in getting a "ZERO" for that question. Good Luck!!!

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CONFIDENTIALITY AGREEMENT & STATEMENT OF HONESTY

I confirm that I will keep the content of this assignment/examination confidential.

I confirm that I have not received any unauthorized assistance in preparing for or doing examination. I confirm knowing that a mark of 0 may be assigned for copied work.

For Online Test/Examination in Comp 3150 Winter 2020: (additional rules to be observed):

1. I confirm that I agree to write this final examination as a closed book examination.
2. I confirm that I am the student with the name and student id signed below.
3. I confirm that I agree to not send email, chat, text or talk in any way to people other than the instructor or proctoring GA of this course during this examination.
4. I confirm that I agree to not engage in copying or cheating during this online examination.

Student Signature

Student Name (please print)

Student I.D. Number

Date

For marking purposes only (This part not to be filled by students)

Question	Mark
Section A (30 marks for 15 multiple choice questions)	/30
Section B	
Que 1 (20 marks)	/20
Que 2 (15 marks)	/20
Que 3 (20 marks)	/20
Que 4 (15 marks)	/10
Total	/100

Section A

30 marks for 15 Multiple Choice Questions. Each question in this section is worth 2 marks.

Fig. 1: Music Database: For music performed by musicians: Note genre means class

MUSICIAN

<u>Ssn</u>	Mname	Genre	Age
11	James Adam	Pop	24
22	Mary Kata	Pop	36
33	Pat Smith	Country	19
44	Cathy Good	Hip Hop	70

SONG

<u>Stitle</u>	Author	Minuteslong
Hello World	21 Savage	25
Happy All	21 Savage	32
Love Sosa	Chief Keef	40

PERFORMS

<u>Ssn</u>	<u>Stitle</u>	Hours
11	Hello World	20
11	Happy All	10
22	Hello World	10
22	Happy All	12
33	Hello World	12
33	Happy All	18
33	Love Sosa	15

Use the MUSIC database of Fig 1 to answer the following questions.

1. The result of the relational algebra query
 $(\pi_{Mname}(\sigma_{Stitle='Hello World'}(Performs \bowtie Musician)))$ is:

- A. James Adam
- B. Mary Kata
- C. James Adam
Mary Kata
- D. James Adam
Mary Kata
Pat Smith
- E. None of the above

2. The SQL query to retrieve the average number of hours performed on songs by musicians from each genre is ____.
- A. SELECT genre, avg(hours) from performs group by genre;
 - B. SELECT p.ssn, m.genre, sum(p.hours) from performs p, musician m where p.ssn=m.ssn group by p.ssn, m.genre;
 - C. SELECT m.genre, avg(p.hours) from performs p, musician m where p.ssn=m.ssn group by m.genre;
 - D. All of the above
 - E. None of the above
3. In SQL, to get the song titles of songs performed on by musicians from Pop or Country genres, use?
- A. SELECT Stitle FROM Song where genre = 'Pop' or genre = 'Country';
 - B. SELECT Stitle FROM Song, Performs where genre = 'Pop' or genre = 'Country';
 - C. SELECT Stitle FROM Song, Performs, Musician where genre = 'Pop' or genre = 'Country';
 - D. All of the above
 - E. None of the above
4. The SQL query for retrieving and presenting unique (not duplicates) musician names with their age, where only age is in sorted descending order (that is big to small order) is:
- A. SELECT distinct(Mname), age FROM MUSICIAN ORDER BY age;
 - B. SELECT distinct(Mname), age FROM MUSICIAN ORDER BY mname desc;
 - C. SELECT distinct(Mname), age FROM MUSICIAN ORDER BY age desc;
 - D. SELECT distinct(Mname), age FROM MUSICIAN ORDER BY mname asc;
 - E. None of the above
5. With SQL, how can you delete the records where the Hours is 18 from the PERFORMS Table?
- A. DELETE ROW Hours = 18 FROM PERFORMS;
 - B. DELETE Hours = 18 FROM PERFORMS;
 - C. DELETE FROM PERFORMS WHERE Hours = 18;
 - D. REMOVE Hours = 18 FROM PERFORMS;
 - E. None of the above

Fig. 1: Music Database: For music performed by musicians: Note genre means class

MUSICIAN

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33	Pat Smith	Country	19
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SONG

<u>Stitle</u>	Author	Minuteslong
Hello World	21 Savage	25
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PERFORMS

<u>Ssn</u>	<u>Stitle</u>	Hours
11	Hello World	20
11	Happy All	10
22	Hello World	10
22	Happy All	12
33	Hello World	12
33	Happy All	18
33	Love Sosa	15

Note: MUSIC database of Fig 1 replicated here just for ease of reference.

6. What is retrieved by the following SQL query from Music database?

Select m.Mname from Musician m where NOT EXISTS
(select * from Performs p where p.Ssn = m.Ssn and p.Stitle = 'Hello World');

- A. James Adam
- B. Mary Kata
- C. Pat Smith
- D. Cathy Good
- E. None of the above

7. From Fig. 1, the Relational algebra query to retrieve Stitle of all songs performed on by 'Mary Kata' is ____.

- A. $\pi_{Stitle} (\sigma_{Mname='Mary Kata'} (Musician))$
- B. $\pi_{Stitle} (\sigma_{Mname='Mary Kata'} (Performs))$
- C. $\pi_{Stitle} (\sigma_{Mname='Mary Kata'} (Song))$
- D. All of the above
- E. None of the above

8. The result of the following relational calculus query on Fig. 1 database is:
 $\{m.Ssn, m.Mname \mid \text{Musician}(m) \text{ and } m.age < 40\}$

A.

Mary Kata
Pat Smith

B.

11	James Adam
22	Mary Kata
33	Pat Smith

C.

22	Mary Kata
33	Pat Smith

D.

44	Cathy Good
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E. None of the above

9. The following natural join relational algebra query ($\text{Performs} \bowtie \text{Musician}$) is equivalent to:

- A. $\text{Performs} \cup \text{Musician}$.
- B. $\text{Performs} \times \text{Musician}$.
- C. $\text{Performs} \cap \text{Musician}$
- D. $\sigma_{\text{Performs.Ssn}=\text{Musician.Ssn}}(\text{Performs} \times \text{Musician})$.
- E. None of the above

10. Which of the following relational algebra operators require that its operands be type compatible?

- A. Cartesian product
- B. Set difference
- C. Natural join
- D. All of the above
- E. None of the above

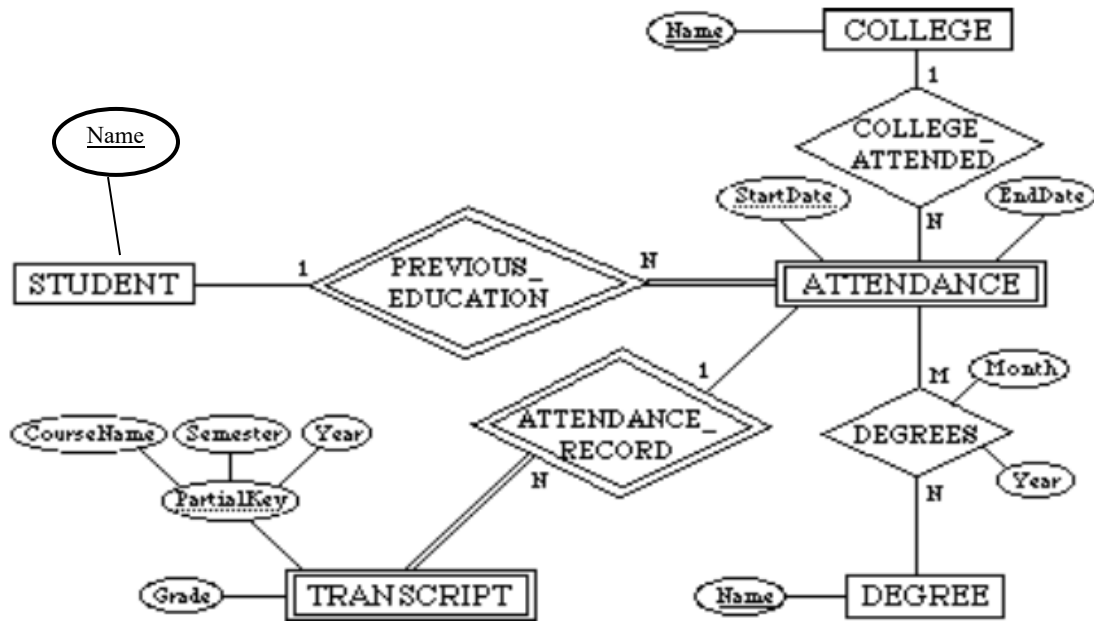
11. With B-tree Indexes, all data nodes are stored on the leaf nodes.

- A. TRUE
- B. FALSE
- C. NOT APPLICABLE
- D. All of the above
- E. None of the above

12. A secondary index _____ :
- A. Has its index field the same as the ordered key field in the primary data file.
 - B. Is the same as a primary index.
 - C. Has to be a single level index.
 - D. All of the above
 - E. None of the above
13. While inserting records in the database, possible integrity constraint violation is:
- A. Domain constraint.
 - B. Entity constraint.
 - C. Referential integrity constraint.
 - D. All of the above
 - E. None of the above
14. The Music database of Fig. 1 is not in 3NF.
- A. FALSE.
 - B. TRUE.
 - C. MAY BE.
 - D. All of the above
 - E. None of the above
15. The Music database of Fig. 1 is in BCNF.
- A. FALSE.
 - B. TRUE.
 - C. MAY BE.
 - D. All of the above
 - E. None of the above

Section B (70 marks):
This section has 4 questions:

Fig. 2: The ER model Diagram for a UNIVERSITY database Database



Consider the ER diagram of Fig 2 for a UNIVERSITY database that is used to keep track of students' transcripts. This is similar but not identical to the database shown in the Figure 1.2 in the book used in class.

Question 1: (total marks: 20 for 10 + 6 + 4)

- Define a relational database schema representation for this database system ER model, clearly defining the table schemas for 6 of the 9 tables for entities and relationships, with their attributes, identifying their primary keys and foreign key attributes constraints. Include in your 6 the two tables **College_Attended** and **Attendance_Record**. (10 marks)
- Write all the SQL instructions to create the 6 of the database tables including the two tables **College_Attended** and **Attendance_Record** with all their primary and foreign key constraints specified. (6 marks)
- Write an SQL query for the English query "Print the student name, the college name, the start date and end date of a student named Pete Adams". (4 marks)

Solution for Question 1 (20 marks for all of a to c)

Sub questions	Answers
a. Relation DB for Fig 2. For both Entities and attributes (5 marks) in ER And Relationships and attributes (5 marks) in ER	
b. creating the 6 tables (6 marks)	

c. SQL query (4 marks)	

Question 2: (total marks = 20 for 5 + 5 + 5 + 5)

- (i) Consider the universal relation $R = \{\underline{A}, \underline{B}, C, D, E, F\}$ and the set of functional dependencies in this database is:

FD1: $\{A, B\} \rightarrow \{C, D, E, F\}$,

FD2: $\{D\} \rightarrow \{E, F\}$.

Given that the key for R is $\{A, B\}$,

- a. Is the relation R in 3NF? Explain your answer using functional dependencies. (2.5 marks)

- b. Decompose R into 3NF relations specifying the primary keys and functional dependencies that exist in each of the relations and thus, in the database. (5 marks)

(ii) Given the STUDENT record relation schema:

STUDENT(Studentid, Courseid, Studentname, Course_grade)

Assume that a student may take multiple courses and that percentage grade is entered. The primary key is (Studentid, Courseid). Other dependencies may exist from the meaning of this domain.

- a. Is the relation STUDENT in 3NF? Explain your answer using functional dependencies. (2.5 marks)
- b. Normalize the relation STUDENT into 3NF set of relations if not in 3NF, showing your new normalized database with its functional dependencies. (5 marks)

Que 2 (20 marks for all of (i) a, b and (ii) a, b)

(i) (a) Is R in 3NF ? (5 marks)	
(b) Decompose R into 3NF. Show FDs. (5 marks)	

(ii) (a) Is STUDENT in 3NF? (5 marks)	
(b) Normalize STUDENT. Show FDs. (5 marks)	

Fig 3: An Instance of a COMPANY relational Database

WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

<u>Pname</u>	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMPLOYEE

<u>Fname</u>	<u>Minit</u>	<u>Lname</u>	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

<u>Dname</u>	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Question 3: (total marks = 20 for 10 + 10)

Using the COMPANY database of Fig 3, answer the following questions by expressing the queries in:

- (i) SQL,
- (ii) Relational algebra,
- (iii) Relational Calculus as well as showing the,
- (iv) result of each query. Note: specify any needed aggregate function or group by attributes within your relational calculus and algebra queries appropriately.

(a) Retrieve the names of male employees who work on project with id 2 and who also have a male dependent. (10 marks)

(b) For each project, list the project name and the minimum hours per week (by all employees) spent on that project. For example, the minimum hours on project 30 (Newbenefits) is 5. (10 marks)

** Some symbols that may be useful for copying are: σ , π , $\leftarrow$, $\bowtie$

Que 3 (10 marks for each of a and b)

(a) Query in SQL, Relational Algebra, Relational Calculus, Result (10 marks)	(a) Retrieve the names of male employees who work on project with id 2 and who also have a male dependent. <u>SQL:</u> <u>Relation Algebra:</u> <u>Relational Calculus:</u>
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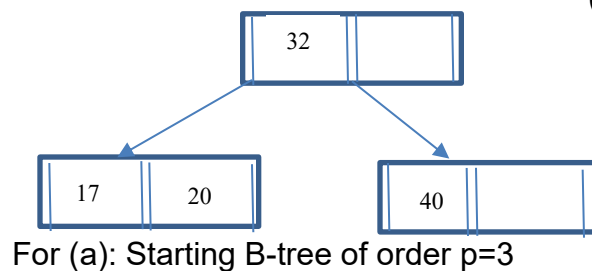
	<u>Query Result:</u>
(b) Query in SQL, Relational Algebra, Relational Calculus, Result (10 marks)	(b) For each project, list the project name and the minimum hours per week (by all employees) spent on that project. <u>Relational Algebra:</u> <u>Result:</u>

Question 4 : (total marks = 10 for 5 + 5)

(a). Given the following B-tree of order $p=3$ which has already inserted into it the values 40, 17, 32, 20, you are required to insert into it, the sequence of ids 31, 99, 43 and show:

- (i) the steps for inserting the given sequence of unique ids for some database records of a relation, and
- (ii) the final B-tree index structure of order $p=3$ (or with maximum of 3 pointers) after the insertions. B-tree is used for speeding up retrieval of records from their primary data file stored on disk.

Show the steps through arriving at your final tree before giving your final tree.
(5 marks)



Solution (a) (5marks)

Give each step of the insertion of 31, 99, 43 into the Starting B-tree with 40, 17, 32, 20 already inserted given above.

Step 1: insert 31 into the above B-tree to get:

Step 2: insert 99 into the above B-tree from step 1 to get:

Step 3: insert 43 into the above B-tree from step 2 to get:

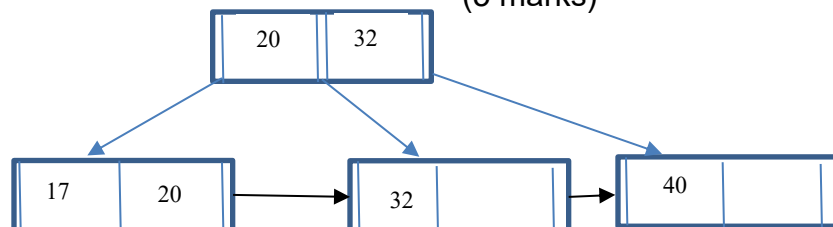
This is the final B-tree after inserting 31, 99, 43 into B-tree with 40, 17, 32, 20.

(b). Given the following B+-tree of order $p=3$ (and p -leaf of 2), which has already inserted into it the values 40, 17, 32, 20, you are required to insert into it, the sequence of ids 31, 99, 43 and show:

- (i) the steps for inserting the given sequence of unique ids for some database records of a relation into the B+-tree, and
- (ii) the final B+-tree index structure of order $p=3$ (or with maximum of 3 pointers) after the insertions. B+-tree is used for speeding up retrieval of records from their primary data file stored on disk.

Show the steps through arriving at your final tree before giving your final tree.

(5 marks)



For (b): Starting B+-tree of order $p=3$ (and p -leaf = 2)

Solution (b) (5marks)

Give each step of the insertion of 31, 99, 43 into the Starting B+-tree above with 40, 17, 32, 20 already inserted.

Step 1: The following shows the B+-tree after insertion of 31.

Step 2: The following shows the B+-tree after insertion of 99.

Step 3: The following shows the B+-tree after insertion of 43.

This is the final B+-tree after inserting 31, 99, 43 into the B+-tree with 40, 17, 32, 20 already inserted.